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ABSTRACT

This research focuses on improving the Blade Element Momentum Theory (BEMT) for low advance ratio conditions, particularly for Vertical Takeoff and Landing (VTOL) aircraft during hover and takeoff. Traditional BEMT solvers often assume a simplified geometry with a single airfoil distributed along the entire blade, which fails to capture the aerodynamic complexity of commercially optimized propellers that utilize varying cross-sections. To address the limitations of single-airfoil methodologies, this study develops a computational framework capable of evaluating multi-profile propeller geometries. The implemented algorithm dynamically applies the XFOIL solver to generate and obtain accurate aerodynamic polar data across different airfoil sections along the blade span. Subsequently, the BEMT methodology is applied to the propeller, quantifying its multi-profile characteristics by assigning the correct polar data to each specific blade element. This dynamic cross-section modeling is integrated alongside essential tridimensional correction factors, including tip loss and hub loss factors, to account for finite blade effects and rotational flow. The computed thrust, torque, and power are evaluated across various operating conditions and compared against established experimental propeller data. By refining aerodynamic coefficient estimation through dynamic multi-profile modeling and appropriate loss corrections, this research quantifies the improvements in BEMT performance and provides a more robust, accurate framework for designing and analyzing complex propeller aerodynamics in VTOL applications.

1. Introduction

The rapid development of electric Vertical Take-Off and Landing (eVTOL) aircraft and Distributed Electric Propulsion (DEP) systems has created a critical need for rapid, yet highly accurate, conceptual design tools [Kim et al. 2025]. While Computational Fluid Dynamics (CFD) provides high-fidelity aerodynamic analysis, its immense computational cost makes it impractical for the extensive iterative processes required during the early stages of rotor design and vehicle sizing. Consequently, semi-empirical methods such as the Blade Element Momentum Theory (BEMT) have remained the industry standard due to their computational efficiency and adaptability for dynamic flight simulations (May, Milz and Looye, 2023) .

Despite its widespread adoption, classical BEMT struggles to accurately predict propeller performance under the complex aerodynamic conditions typical of VTOL operations, particularly during hover and takeoff at low advance ratios doe to the induced post-stall angles of attack (AOA). A primary structural limitation of traditional BEMT solvers is the assumption of a simplified, uniform geometry that utilizes a single airfoil distributed across the entire blade span. This simplification starkly contrasts with the geometric reality of modern, commercially optimized propellers—such as the APC Thin Electric family. To manage the vast variations in local Reynolds numbers and prevent structural or aerodynamic inefficiencies, these rotors frequently employ transitional, multi-profile geometries, shifting from profiles like the E63 near the root to a NACA 4412 toward the tip (Danial, 2015) .

Applying a single aerodynamic polar to such complex geometries leads to significant uncertainties in performance estimation. While some modern solvers attempt to bypass this by utilizing pre-calculated static databases, there remains a critical gap in dynamically generating and assigning the correct aerodynamic data to specific blade sections while simultaneously correcting for severe three-dimensional flow phenomena, such as wake rotation and pressure leaks at the blade extremities [Sanangelantoni 2023].

To overcome the limitations of single-airfoil methodologies, this research proposes an advanced computational framework that significantly enhances BEMT for low advance ratio conditions. The core innovation of this study is the development of an algorithm that dynamically applies the XFOIL solver to map, generate, and extract precise aerodynamic polar data across the varying cross-sections of a multi-profile propeller. By dynamically quantifying the blade's multi-profile characteristics, the framework assigns the exact aerodynamic polar to each specific radial element.

Furthermore, because traditional BEMT inherently assumes an infinite number of blades, this dynamic cross-section modeling is strictly coupled with essential tridimensional flow corrections. The framework integrates Prandtl's

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tip loss and hub loss factors to account for finite blade effects, properly adjusting the induced velocities at the rotor's inner and outer bounds [Sanangelantoni 2023]. By refining the aerodynamic coefficient estimation through dynamic multi-profile generation and rigorous 3D loss corrections, this methodology effectively computes total thrust, torque, and power. Ultimately, this research provides a robust, highly accurate framework for evaluating complex propeller geometries, offering a vital tool for the optimization and conceptual design of advanced eVTOL propulsion systems.

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Applying a single aerodynamic polar to complex propeller geometries leads to significant uncertainties in performance estimation. Historically, this simplification was adopted not only to reduce computational costs but also due to manufacturing constraints; as noted by Danial (2015), optimizing multiple profiles through cross-sectional interpolation often resulted in irregular designs that were structurally difficult to produce using traditional methodologies.

Despite the modern necessity for modeling complex commercial propellers, recent BEMT solvers still face significant barriers in multi-profile implementation. Sanangelantoni (2023), for instance, structured the logic for multi-polar use in a MATLAB-based BEMT algorithm but identified the formulation's complexity as a primary limitation, leaving the integration of multi-profile calculations strictly for future work. Recognizing these gaps, the current state of the art has shifted toward radial variation modeling. The ExBEMT solver (Konuk, 2024) represents this trend, employing a "dynamic section detection" system and utilizing neural networks—specifically NeuralFoil—alongside pre-calculated databases to ensure high fidelity during the low-advance-ratio flights typical of eVTOL operations.

To advance beyond these existing methodologies, this research proposes a computational framework that dynamically bridges the gap between section-specific geometry and aerodynamic response. The core innovation is an algorithm that dynamically invokes the XFOIL solver to map, generate, and extract precise aerodynamic polar data across the varying cross-sections of a multi-profile propeller. Unlike static database approaches, this framework quantifies the blade's unique characteristics in real-time, assigning the exact aerodynamic polar to each specific radial element.

Furthermore, because traditional BEMT inherently assumes an infinite number of blades [Branlard 2017], this dynamic modeling is coupled with essential tridimensional flow corrections. The framework integrates Prandtl's tip and hub loss factors [Glauert 1935] to account for finite blade effects, properly adjusting induced velocities at the rotor's boundaries [Hansen 2008]. By refining coefficient estimation through dynamic generation and rigorous 3D corrections, this methodology effectively computes total thrust, torque, and power for complex geometries.

The remainder of this article is organized as follows: Section 2 details the mathematical formulation of the BEMT and the integration of the XFOIL-based dynamic mapping; Section 3 describes the computational implementation and the validation cases; Section 4 presents the results for various eVTOL propeller geometries; and Section 5 concludes with final remarks on the framework's utility for conceptual design and optimization.

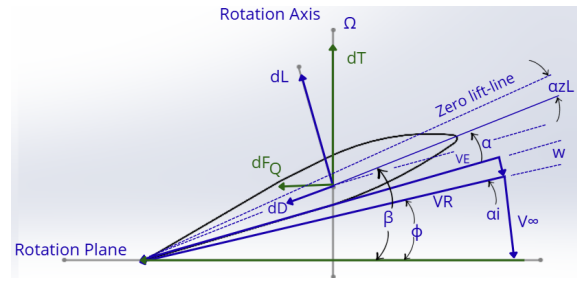

Figure 1: BEMT.

2. Theoretical Foundation

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2.1. BEMT

The methodology employed in this research utilizes the Blade Element Momentum Theory (BEMT), which integrates classical Blade Element Theory (BET) with Momentum Theory. The objective of this implementation is to determine the induced velocity (w) and the induced angle of attack (α_i). Within the BET framework, propeller performance is estimated by discretizing each blade into several segments, or blade elements, which are analyzed as independent two-dimensional airfoils. This approach allows for the application of varying geometric conditions at each radial distance (r) from the hub—most notably the chord ($c(r)$) and pitch angle (β)—which exhibit significant variations along the blade span. For clarity, the forces and velocity vectors acting on a two-dimensional blade element are illustrated in Fig. 2.

The differential Lift and Drag forces can be written such as in Eq. (1) and Eq. (2), the helix angle ϕ is calculated on the Eq. (3), from the forward airspeed to be simulated V_∞, and the elemental induced velocity w. The effective resultant velocity V_E is estimated as shown in Eq. (4), and α_i,the angle-of-attack induced by w is estimated in Eq. (5).

The following equations describe the differential lift, drag, velocities, and induced angles along the blade span.

$$dL = \frac{1}{2}\rho V_E^2 c\,(r)\,Cl\,dr \tag{1}$$

$$dD = \frac{1}{2}\rho V_E^2 c\,(r)\,Cd\,dr \tag{2}$$

$$\phi = \tan^{-1}\left(\frac{V_\infty}{w\,r}\right) \tag{3}$$

$$V_E = \sqrt{\left(V_\infty + w\right)^2 + (\Omega r)^2} \tag{4}$$

$$\alpha_i = \sin^{-1}\left(\frac{w}{VR}\right) = \sin^{-1}\left(\frac{w}{\sqrt{V_\infty^2 + (\Omega r)^2}}\right) \tag{5}$$

Thus, the differential thrust, torque, and power can be calculated as presented in Eq. (6) Eq. (7) Eq. (8), respectively. The total thrust, torque, and power are calculated by integrating these differential equations from the hub to the tip of the propeller, and number of blades (N_B).

$$dT = dL\cos\left(\phi + \alpha_i\right) - dD\sin\left(\phi + \alpha_i\right) \tag{6}$$

$$dQ = r\left[dL\sin\left(\phi + \alpha_i\right) + dD\cos\left(\phi + \alpha_i\right)\right] \tag{7}$$

$$dP = \Omega\, dQ \tag{8}$$

Since the induced angle of attack α_i is necessary to select the correct C_l and C_d from the dataset of each airfoil section, this is accomplished using the Momentum Theory to determine the induced velocity by applying the Newton–Raphson iterative scheme to obtain the root w of Eq.(9).

$$f(w) = \left(\frac{8\pi r}{N B\, c(r)}\right) w - \sqrt{1 + \left(\frac{\Omega^2 r^2}{V_\infty + w}\right)}\left[C_l\left(\Omega r\right) - C_d\left(V_\infty + w\right)\right] = N_B F_P \int_{R_{\text{hub}}}^{R} \Omega r\left[dL\sin(\phi + \alpha_i) + dD\cos(\phi + \alpha_i)\right] \tag{9}$$

This integration of Momentum Theory to resolve the flow field deficiencies in the classical Blade Element Theory (BET) enables the robust application of BEMT for the static and low-speed conditions investigated in this research.

2.2. Prandtl's Tip and Hub Loss Corrections

Near the tip of any lifting surface, the local lift coefficients drop to zero. Standard Blade Element Theory (BET) does not account for this, which introduces errors in the predictions. The application of the correction deviser by Prandtl (1919) solves this and also extends to account for hub corrections, where lift similarly goes to zero. The correction can be applied using this three-step process:

Calculating the tip correction parameter P_{tip} with Eq. (10), the tip correction factor F_{tip} with Eq. (11), then the hub correction parameter P_{hub} in Eq. (12) and the hub correction factor F_{hub} at Eq. (13) The overall correction factor is ilustrated in Eq. (14) and is the combination of the F_{tip} and the F_{hub}, and varies with r.

$$P_{\text{tip}} = \frac{N_B}{2}\frac{R - r}{r\sin\phi} \tag{10}$$

$$F_{\text{tip}} = \frac{2}{\pi}\cos^{-1} e^{-P_{\text{tip}}} \tag{11}$$

$$P_{\text{hub}} = \frac{N_B}{2}\frac{r - R_{\text{hub}}}{r\sin\phi} \tag{12}$$

$$F_{\text{hub}} = \frac{2}{\pi}\cos^{-1} e^{-P_{\text{hub}}} \tag{13}$$

$$F_P = F_{\text{hub}}\, F_{\text{tip}} \tag{14}$$

It is then used to modify the BET equations for thrust, torque, and power as follows:

$$T = N_B F_P \int_{R_{\text{hub}}}^{R}\left[dL\cos(\phi + \alpha_i) + dD\sin(\phi + \alpha_i)\right] \tag{15}$$

$$Q = N_B F_P \int_{R_{\text{hub}}}^{R} r\left[dL\sin(\phi + \alpha_i) + dD\cos(\phi + \alpha_i)\right] \tag{16}$$

$$P = N_B F_P \int_{R_{\text{hub}}}^{R} \Omega r\left[dL\sin(\phi + \alpha_i) + dD\cos(\phi + \alpha_i)\right] \tag{17}$$

2.3. HUB FLOW CORRECTION

In a Blade Element Momentum Theory (BEMT) framework, the induced velocities and aerodynamic forces are typically computed assuming two-dimensional sectional characteristics. However, in the inner radial region of the blade, close to the hub, the flow is strongly affected by three-dimensional rotational effects. These effects modify the boundary layer behavior, particularly delaying flow separation and increasing the effective lift coefficient compared to standard 2-D data, Du and Selig (1998). Neglecting these phenomena leads to significant underprediction of the aerodynamic loads generated in the inboard region of the propeller.

The rotation of the blade introduces centrifugal and Coriolis forces in the boundary layer equations. These terms stabilize the flow, delaying laminar separation, as captured in the separation factor k_s:

$$k_s = f\left(\frac{c}{r}, \Lambda\right), \tag{18}$$

Table 1

where c/r is the chord-to-radius ratio and $\Lambda = \frac{\Omega r}{V_\infty}$ is the local non-dimensional rotation speed. Increased rotation causes the separation point to move downstream, increasing the maximum lift coefficient and reducing drag relative to non-rotating conditions.

For a 2-D section, the stall onset occurs when $k_s = k_{s,2D} \approx 0.1267$. Under rotating conditions, k_s increases:

$$\Delta k_s = k_s - k_{s,2D} > 0, \tag{19}$$

Indicating that the stall is delayed at inboard stations. This effect is strongest where c/r is large, near the hub.

To account for this in a BEMT model, hub flow corrections modify the 2-D lift and drag data as

$$C_{l,3D} = C_{l,2D} + f_l(c/r, \Lambda)\left(C_{l,p} - C_{l,2D}\right), \tag{20}$$

$$C_{d,3D} = C_{d,2D} - f_d(c/r, \Lambda)\left(C_{d,2D} - C_{d,0}\right), \tag{21}$$

Where f_l and f_d are empirical correction factors, and $C_{l,p}$ is the potential lift. These corrections ensure that the increased lift and reduced drag observed in rotating flows are represented in the inboard blade elements improving the accuracy of thrust and torque predictions near the hub.

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2.4. MORPH AIRFOIL

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2.5. POLAR GENERATION montgomerie + xfoil

Here are two sample references: See **?**. Also refer **??**. More citations are here (**??**).

3. METODOLOGY

3.1. DATA EXTRACT

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3.2. DIAGRAM OF OPERATIONS

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4. resultados

5. conclusão

6.

7.

CRediT authorship contribution statement

P.A. Madureira: . **Odenir de almeida:** . **Filipi rizze:** .

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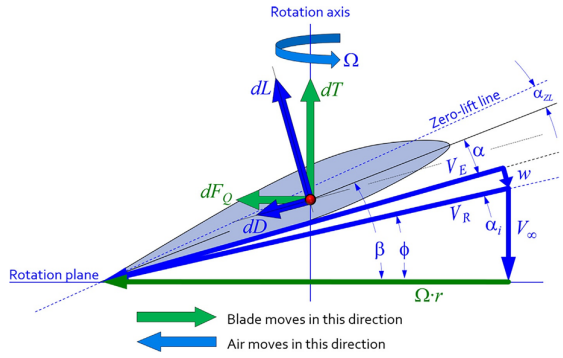

Figure 2: Definition of forces, angles, and velocity for the propeller blade.(Gudmundsson, 2022)

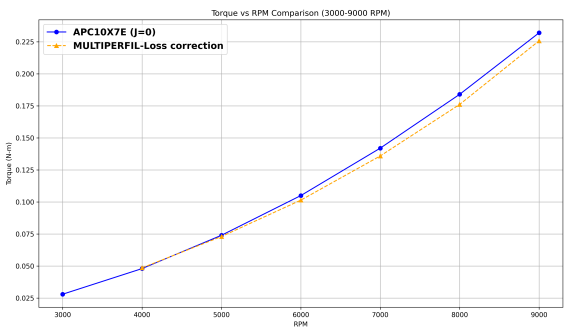

Figure 3: results 10x7e. (See also Table 1).

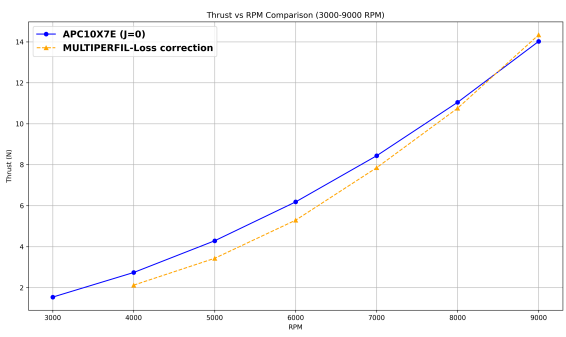

Figure 4: results 10x7e. (See also Table 1).

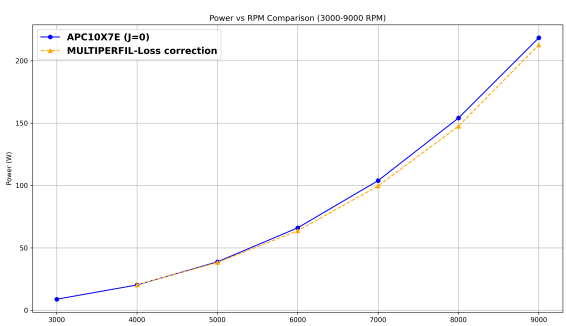

Figure 5: results 10x7e. (See also Table 1).

Figure 6: